

Market Analysis: AIIA Clinical Trials Dashboard

Team MUGIWARA'S · Team ID 128194 · Problem Statement PS26046

Smart India Hackathon 2026 · Theme: MedTech / BioTech / HealthTech · Category: Software

1. Purpose and method

This document sizes the opportunity for a cloud-based, GCP-compliant Clinical Trial Management System (CTMS) built for Ayurveda research, and derives **TAM, SAM and SOM** using transparent arithmetic from published registry data.

Method note: every figure below is either (a) sourced and cited, (b) derived arithmetically from a sourced figure with the derivation shown, or (c) an explicitly labelled planning assumption. Category (c) items are marked **[assumption]** and must be re-sourced before being used in a funding or procurement context.

2. The market gap

Ayurveda clinical research in India runs at meaningful volume but on spreadsheets and email. Global CTMS products embed ICH-GCP pharma workflows and do not encode ASU-specific regulation (GCP-ASU, pharmacovigilance reporting under Ayush Suraksha), and their indicative licensing cost of ₹10–15 lakh (~\$12k–18k) is out of reach for most AYUSH institutions. The result is a genuine unmet need rather than a feature gap in an existing category.

3. Sourced base data

Data point	Value	Source
India clinical trials market by 2034	₹38,900 cr	IMARC, as cited in project materials
Ayurveda trials registered in CTRI, 2018–20	1,392	Clinical Trials Registry – India (ctri.nic.in)
Total trials on CTRI since 2007	107,000+	Clinical Trials Registry – India (ctri.nic.in)
Regulatory framework	GCP-ASU / ICMR-NETH; NDCT Rules 2019	cdsr.nic.in; indiacode.nic.in
Pharmacovigilance framework	Ayush Suraksha / NPvCC	suraksha.ayush.gov.in
Ayurveda mandate	Ministry of Ayush, incl. AIIA	aiia.gov.in

Derived annual run-rate: $1,392 \text{ Ayurveda trials} \div 3 \text{ years (2018–20)} \approx \mathbf{464 \text{ Ayurveda trial registrations per year}}$. This is the central driver of the SAM calculation below.

4. TAM — Total Addressable Market

Definition: the Indian clinical-trials market that a compliant trials-management system could serve.

Value: ₹38,900 cr by 2034 (IMARC).

This figure covers the whole Indian clinical-trials ecosystem, of which trial-management software is only a component. It is used here as an **outer bound** on the opportunity, not as the addressable software revenue. A software-only TAM would be materially smaller; deriving it would require a licensed software-market dataset that this project does not currently hold. Stating the ecosystem figure avoids overstating a number the team cannot defend.

5. SAM — Serviceable Addressable Market

Definition: the Ayurveda-focused trials in India that require GCP-ASU-compliant trial management and would realistically buy software of this type.

Two independent framings, deliberately presented together because each is weak alone:

5.1 Volume framing (bottom-up)

- Ayurveda trial registrations: ≈ 464 per year (derived, §3).
- Assume each trial runs 2–3 years **[assumption]** → an active Ayurveda trial population of roughly **930–1,400 trials at any time**.
- Not every active trial buys software; assume **20–40%** are in institutions able and willing to procure **[assumption]** → $\approx$ **190–560 addressable active trials**.

5.2 Spend framing (top-down)

- At the proposed per-study price point (\$7), 190–560 addressable active trials at ₹15,000–30,000 per study per year gives a SAM of roughly **₹0.3–1.7 crore per year**. The wide band is honest: it reflects three stacked assumptions, not precision.

5.3 SAM conclusion

Indicative SAM: ₹0.3–1.7 crore per year of software subscription spend in Indian Ayurveda trials.

Treat as a planning band, not a forecast. The key point is that this is a *niche but real, registry-verified* demand — 1,392 Ayurveda trials were registered in three years — not a hypothetical one.

6. SOM — Serviceable Obtainable Market

Definition: what the team can realistically capture in the first 18–24 months.

The beachhead is deliberately narrow: institutions already mandated to run Ayurveda trials, reachable through the AYUSH network and AIIA relationships, where adoption is driven by principal investigators rather than by procurement.

Bottom-up SOM model:

Input	Value	Basis
Target AYUSH institutions (colleges, hospitals, research institutes)	*N*	[to be sourced] from Ministry of Ayush / AYUSH college listings
Capture rate in 18–24 months	8–15% [assumption]	Early-adopter rate for a zero-licence-fee, PI-led product
Institution licence price	₹1,50,000–3,00,000 per year	\$7

	[assumption]	
Per-study add-on	₹15,000–30,000 per study per year [assumption]	\$7

At N = 30 institutions and 12% capture, that is $\approx$ 3–4 paying institutions, or roughly **₹4.5–12 lakh of annual recurring revenue** in year one. The SOM is therefore small in absolute terms and should be presented as a credibility-of-revenue exercise, not a scale case. The strategic value is a defensible reference position in a category that does not yet exist for ASU trials.

Sourcing gap to close before submission: the AYUSH institute count (N) is the single input that most changes the SOM. It should be taken from official Ministry of Ayush institution listings rather than estimated.

7. Pricing position

Proposed model (developed in full in the *Business Perspective* document):

Offering	Price	Rationale
Institution licence	₹1,50,000–3,00,000 / year	Anchored well below the ₹10–15 lakh global-CTMS licence
Per active study	₹15,000–30,000 / study / year	Matches per-study CTMS norms, priced for AYUSH budgets
Setup and onboarding	₹50,000 one-time	Migration, configuration, role mapping
Guided training	Included in setup; ₹1,000 / person / day thereafter	Removes the adoption barrier
Grant-funded pilots	Discounted or deferred	Pilot-to-paid conversion is the primary funnel

Important correction to the pitch deck: the presentation currently presents ₹2,500–6,000 per month as the "with dashboard" software cost and derives a –95% saving from it. That figure is the **cost to run** the serverless stack, not a price. Presenting a hosting cost as the replacement price understates achievable revenue and would not survive scrutiny from a judge or an institutional buyer. The commercial model above should replace it.

8. Competitive landscape

Competitor type	Strength	Gap exploited
Global commercial CTMS/EDC (ICH-GCP pharma)	Mature, validated, well funded	No ASU regulation; ₹10–15 lakh licence; heavy implementation
India CRO / CRO-hosted trials management	Domain expertise, existing client base	Services, not a product; per-trial cost; no reusable audit trail
Hospital information systems	Already installed at sites	Built for clinical care, not trials; no GCP-ASU rule engine or CTRI awareness
Spreadsheets and email	Free, universally adopted	No audit trail, no safety

		escalation, no compliance guarantee
--	--	-------------------------------------

The team's differentiator is not feature count — it is that **Ayush regulation is encoded in the software**, so the audit trail is produced automatically rather than assembled afterwards.

9. Risks to the market case

1. **Registry data is not demand.** 464 registrations per year is activity, not willingness to pay. Conversion is unproven.
2. **Procurement inertia.** AYUSH institutions may prefer grant-funded pilots with no ongoing cost, capping ARPU.
3. **No public integrations.** The absence of CTRI/HIS/EDC APIs is both a technical and a commercial constraint.
4. **Single vendor dependence.** A serverless architecture on one cloud provider is a continuity risk.
5. **Adoption risk.** Every revenue assumption is conditional on clinicians changing working practice.

10. Conclusion

The registry data establishes a real, verifiable demand base — 1,392 Ayurveda trials in three years — and the absence of an ASU-aware CTMS establishes a real gap. The resulting opportunity is a **niche but defensible** market: an indicative SAM of ₹0.3–1.7 crore per year, with a first-24-month SOM of ₹4.5–12 lakh. The honest conclusion is that this is a category-defining position rather than a volume play, and that the decisive unknowns are conversion and adoption — not market size. The two figures requiring re-sourcing before external use are the AYUSH institute count (N) and the software-only share of the IMARC market figure.

11. References

- Clinical Trials Registry – India — ctri.nic.in
- IMARC Group, India clinical trials market report (as cited in project materials)
- Ayush Suraksha — suraksha.ayush.gov.in
- GCP / ICMR-National Ethical Guidelines (CDSR) — cdsr.nic.in
- NDCT Rules 2019 — indiacode.nic.in
- HL7 FHIR R4 — hl7.org/fhir
- CDISC — cdisc.org
- All India Institute of Ayurveda, Ministry of Ayush — aiia.gov.in

Status: market sizing for review. Values marked [assumption] are planning inputs, not measured market data, and require validation.